Atmospheric entry

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use energy and geometry to discuss entry heating without claiming a guaranteed outcome.

SUCCESS CRITERIA - I can...

1. Compare kinetic energy with a stated heating requirement.
2. Explain the role of surface area relative to mass.
3. Name missing factors in a survival prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “ablation” and “thermal energy” to explain your answer. Sentence starters are optional.

ablation: loss of surface material during heating or erosion

thermal energy: internal energy associated with microscopic motion/interactions

impact: a collision or strike between objects

01 Warm-up: bring back an earlier idea

A process has $\Delta U = \mathbf{-15\ J}$ and $W_{by} = \mathbf{21\ J}$. **Find Q** and **decide whether** heat enters or leaves.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $K = \frac{1}{2}mv^2$

$Q_{absorbed}$ = fraction absorbed × energy available.

Entry also involves ablation and fragmentation. An energy comparison alone cannot predict survival.

Worked reasoning: A fictional entering object loses **1000 J** of mechanical energy; a supplied calculation assigns **700 J** to heated air and **300 J** to the object. Those shares are values you are told to use, not universal atmospheric-entry fractions.

Your turn: Draw the energy budget and **name one measurement needed** to **test the shares**.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Compare a small** and a large sphere of the same material.
2. Which has more surface area per unit mass?
3. What does that suggest about surface interactions per kilogram?

Sentence starter: Both ... . They differ in ... because ... .



A B

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

An atmospheric-entry simulation starts a hypothetical **10 kg** rock at **15 km/s**. A supplied heating-and-vaporization requirement is **8.6 × 10⁶ J/kg**.
1. **Calculate its kinetic energy**, total requirement and their ratio.
2. **Label possible destinations** for the energy; **explain why the ratio alone** cannot predict survival.

Sentence starter: I predict … . I used … to make this prediction because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

Name at least three additional quantities needed before claiming a particular rock completely ablates or reaches the ground.

Sentence starter: I notice … . This supports … because … .

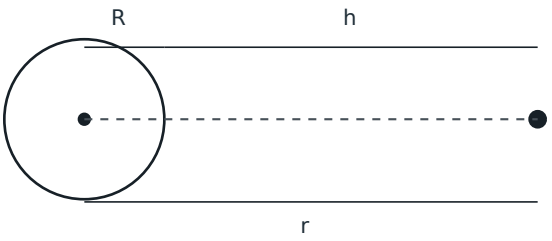
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

For fixed GM, launch radius becomes **16** times as large. **Compare ideal escape speeds** and **name the equation**.
Label the distances or energy levels used in your reasoning.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3

07 Exit check

If radius doubles at fixed density, derive the area-to-mass multiplier and **explain why kinetic-energy-per-mass alone cannot** predict a size threshold for survival. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: I predict … . I used … to make this prediction because … .



Radius r Radius 2r

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **2 kg** sample has **20 kJ** deposited as internal energy. With effective c=**500 J**/(kg·K), estimate ΔT if no phase change or loss occurs.
1. **State why this** is not a survival threshold.
2. **Label energy entering**, leaving or stored on the diagram of the objects you are studying.

Word help: estimate = find an approximate value

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3